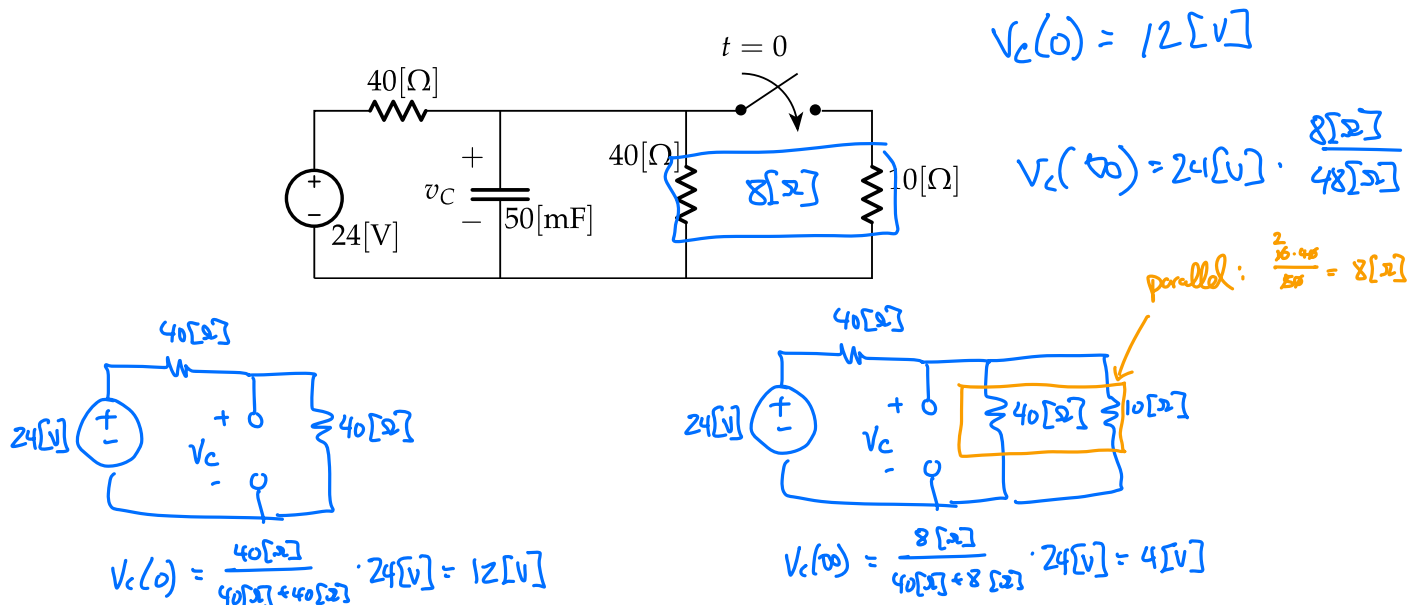


Quiz 7/In-class Assignment

Name: _____ PSID: _____

Problem 1

Determine $v_C(t)$ for $t > 0$, where $v_C(t)$ is the voltage across the capacitor. Assume the switch has been open for a long time before $t = 0$. For your answers, write the values of $v_C(0)$, $v_C(\infty)$, and τ , as well as the expression for $v_C(t)$ in the spaces provided below.



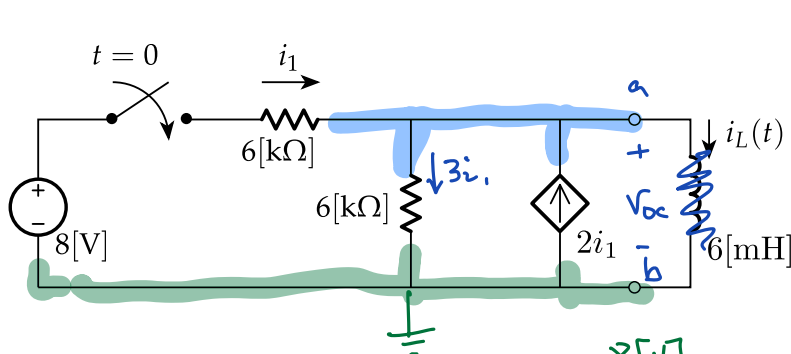
$$\begin{aligned}
 v_C(t) &= v_C(\infty) + (v_C(0) - v_C(\infty))e^{-t/\tau} \\
 &= 4[V] + (12[V] - 4[V])e^{-3t/[s]} \\
 &= 4[V] + 8[V]e^{-3t/[s]}
 \end{aligned}$$

$$\begin{aligned}
 v_C(0) &= 12[V] \\
 v_C(\infty) &= 4[V] \\
 \tau &= \frac{1}{3}[s] \\
 v_C(t) &= 4[V] + 8[V]e^{-3t/[s]}
 \end{aligned}$$

Problem 2

Determine $i_L(t)$ for $t > 0$, where $i_L(t)$ is the current flowing through the inductor. Assume the switch has been open for a long time before $t = 0$. For your answers, write the values of $i_L(0)$, $i_L(\infty)$, and τ , as well as the expression for $i_L(t)$ in the spaces provided below.

Bonus: Sketch a graph of $i_L(t)$ vs. t , labeling axes and units. How many time constants does it take for the value of $i_L(t)$ to get within 5% of the final value. Show this clearly on the graph.



$$i_L(0) = 0$$

$$i_L(\infty) = 3i_1 = 4 \text{ [mA]}$$

$$i_1 = \frac{8 \text{ [V]}}{6 \text{ [k}\Omega]} = \frac{4}{3} \text{ [mA]}$$

$$i_L(t) = i_L(\infty) + (i_L(0) - i_L(\infty))e^{-t/\tau}$$

$$-8 \text{ [V]} + i_1 \cdot 6 \text{ [k}\Omega] + 3i_1 \cdot 6 \text{ [k}\Omega] = 0$$

$$R_{eq} = \frac{V_{oc}}{I_{sc}} = \frac{6 \text{ [V]}}{4 \text{ [mA]}} = \frac{3}{2} \text{ [k}\Omega]$$

$$4i_1 = \frac{8 \text{ [V]}}{6 \text{ [k}\Omega]}$$

$$i_1 = \frac{8}{24} \text{ [mA]}$$

$$V_{oc} = 3i_1 \cdot 6 \text{ [k}\Omega] = \frac{3 \cdot 8 \cdot 6}{24} = 6 \text{ [V]}$$

$$\tau = \frac{L}{R} = \frac{6 \text{ [mH]}}{\frac{3}{2} \text{ [k}\Omega]} = 4 \text{ [\mu s]}$$

$$i_L(0) = 0$$

$$i_L(\infty) = 4 \text{ [mA]}$$

$$\tau = 4 \text{ [\mu s]}$$

$$i_L(t) = 4 \text{ [mA]} (1 - e^{-t/4 \text{ [\mu s]}})$$